# Boishik Barua

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## Summary

Final-year Computer Science and Engineering student with hands-on experience in machine learning, computer vision, NLP, full-stack development, and software engineering. Skilled in Python, PyTorch, TensorFlow, Scikit-learn, JavaScript, Node.js, Express.js, and MySQL. Built AI, research, and web-based projects involving text-to-face generation, hallucination detection, medical image classification, recommendation systems, and multi-role e-commerce platforms.

## Education

**North South University** — Dhaka, Bangladesh
*B.Sc. in Computer Science and Engineering* — *Expected Jun 2026*

- CGPA: 3.57/4.00; Distinction: Cum Laude

**Notre Dame College** — Dhaka, Bangladesh
*Higher Secondary Certificate, Science* — *2020*

- GPA: 5.00/5.00

**Monipur High School** — Dhaka, Bangladesh
*Secondary School Certificate, Science* — *2018*

- GPA: 5.00/5.00

## Technical Skills

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|---|---|
| **Languages** | Python, C, C++, Java, JavaScript, SQL |
| **ML / AI** | Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, CNNs, Vision Transformers, NLP, Knowledge Distillation, Grad-CAM |
| **Web / Backend** | HTML, CSS, JavaScript, Node.js, Express.js, MySQL |
| **Tools** | Git, GitHub, Jupyter Notebook, Google Colab, XAMPP, AutoCAD |
| **Concepts** | Computer Vision, Natural Language Processing, Feature Engineering, Model Evaluation, OOP, Data Structures and Algorithms |

## Projects

**Forensic Text-to-Face Suspect Generation** — 2025 – Present
*Capstone Project*

- Developing a forensic AI system that generates suspect-like facial images from witness text descriptions.
- Applying Stable Diffusion, ControlNet-based conditioning, and geometry-guided generation to improve facial structure consistency.
- Combining natural language descriptions with hierarchical facial geometry cues for controlled image synthesis.

**Bangla Hallucination Benchmark for Summarization** — 2025 – Present
*Research Project*

- Contributing to a benchmark for hallucination detection in Bangla text summarization.
- Experimenting with BanglaBERT, XLM-R, multilingual encoders, and LLM-based baselines for hallucination classification.
- Supporting dataset preparation, model evaluation, and comparative analysis for low-resource NLP.

**Hybrid Feature-Fusion Network with Knowledge Distillation** — 2025 – 2026
*Pattern Recognition Project*

- Built a dental X-ray classification pipeline using ConvNeXt, Swin Transformer, ResNet34, and knowledge distillation.

- Integrated Grad-CAM to visualize model attention and improve interpretability of predictions.
- Designed a hybrid feature-fusion approach for medical image classification.

**Baggy – Multi-Role E-Commerce Platform** 2025
*Full-Stack Web Project*

- Developed a full-stack e-commerce platform using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL.
- Implemented product browsing, cart, wishlist, stock management, order tracking, voucher logic, and payment-status workflows.
- Built database-backed customer, seller, and support role features for multi-user platform functionality.

**BanglarKrishok – Agricultural Expert System Website** 2025
*Software Engineering Project*

- Built a rule-based agricultural recommendation system using Node.js, Express.js, MySQL, JavaScript, and JSON-based crop rules.
- Served as second team lead and contributed to system design, UI mockups, and UML documentation.
- Prepared use case, class, sequence, activity, and state diagrams for software design analysis.

**Movie Recommender System** 2025
*Machine Learning Project*

- Developed and deployed an interactive movie recommendation system using a curated dataset.
- Implemented search-based recommendations using movie title and genre keywords.
- Added a watchlist feature to help users save movies for later viewing.

## Additional Projects

**Currency Converter** 2024
*Java Project*

- Built a Java-based currency conversion application for calculating exchange values across multiple currencies.

**Tic-Tac-Toe Game** 2024
*JavaScript Project*

- Developed an interactive browser-based Tic-Tac-Toe game with turn handling and win/draw condition checks.

**Rock-Paper-Scissors Game** 2024
*JavaScript Project*

- Built a browser-based Rock-Paper-Scissors game with real-time user interaction and score tracking.

## Awards & Certifications

**GCL World 2026 – AI and Data Science Training Program** Apr 2026 – Jul 2026
*Matsuo Iwasawa Lab, University of Tokyo* *Japan*

- Completed intensive training on practical and theoretical implementation of AI, machine learning algorithms, and data science.

**1st Runner-Up – Innovation Challenge Season 19** 2026
*North South University* *Dhaka, Bangladesh*

- Awarded 1st Runner-Up and Best Poster Design for "Face Reconstruction using Hierarchical Geometry Conditioning from Witness Description."

**Intra NSU Junior Programming Contest – Summer 2023** Summer 2023
*North South University* *Dhaka, Bangladesh*

- Ranked 7th and achieved the second-highest number of solved problems.